

Approximate Optimal Strategy for Multiagent System Pursuit–Evasion Game

Zhiqiang Xu , Dengxiu Yu , *Member, IEEE*, Yan-Jun Liu , *Senior Member, IEEE*,
and Zhen Wang , *Fellow, IEEE*

Abstract—In this article, we propose an approximate optimal control strategy for a class of nonlinear multiagent system pursuit–evasion games. Herein, multiple pursuers aim to capture multiple evaders trying to evade capture. Under the competitive framework, agents not only pursue their individual goals but also consider coordination with their teammates to achieve collective objectives. However, maintaining cohesion with teammates in existing distributed control methods has always been a challenge. To enhance team coordination, we employ a graph-theoretic approach to represent the relationships between agents. Based on this, we design a dynamic target graph algorithm to enhance the coordination among pursuers. The approximate optimal strategies for each agent are solved by utilizing the Hamilton–Jacobi–Isaacs equations of the system. As solving these equations becomes computationally intensive in multiagent scenarios, we propose a value-based single network adaptive critic network architecture. In addition, we consider scenarios where the numbers of agents on both sides are inconsistent and address the phenomenon of input saturation. Moreover, we provide sufficient conditions to prove the system’s stability. Finally, simulations conducted in two representative scenarios, multiple-pursuer-one-evader and multiple-pursuer-multiple-evader, demonstrate the effectiveness of our proposed algorithm.

Index Terms—Approximate optimal control, multiagent systems, pursuit–evasion games, reinforcement learning.

I. INTRODUCTION

IN THE past decades, the pursuit–evasion (PE) game has emerged as a prominent and intriguing research topic, drawing significant attention. The pursuers and evaders, which participate in the PE game and possess capabilities in perception, communication, and action, are collectively referred to as agents. The multiagent system pursuit–evasion (MPE) game is a complex extension of the PE game, involving multiple agents. Specifically, the objective of the pursuers is to minimize the distance to their targeted evaders and achieve interception while the evaders aim to evade or delay interception. Unlike evaders who engage in passive avoidance, pursuers possess the flexibility to select different targets as their objectives. The MPE game initially arose in

natural animal hunting scenarios and have since been further developed with the concepts of game theory. The MPE game finds wide engineering applications in areas, such as autonomous aerial vehicles, satellites, mobile robot systems, vicinagearth security, and sports/game strategy [1], [2], [3], [4], [5], [6].

The single-pursuer single-evader game was initially studied based on the differential game framework involving two players, which is a zero-sum game. In most PE games, the pursuer’s objective of minimizing the distance to the evader is represented through value functions. These individual performance indices are then used to define equilibrium points of the game [7]. By differentiating the value functions, Jang and Tomlin [8] achieved PE between aircraft. Jagat and Sinclair [9] based on the state-dependent Riccati equation methods designed a nonlinear control approach for the spacecraft PE game. Later, the PE game was extended to the multiple-pursuer single-evader scenario [10], where pursuers can form certain formations through formation control. When the PE game is extended to the multiple-pursuer multiple-evader scenario, it is referred to as MPE [11]. In [12], different types of pursuers are designed, and the PE relationships are studied based on an N -player nonzero-sum game framework.

In MPE scenarios with multiple agents, the relationships among them become more complex [13], [14], [15]. To handle these relationships, graph-theoretic approaches have been utilized, as demonstrated in [16] and [17] and applied to PE games with linear dynamics in [11]. In [18], the pursuer, acting as a child node, is paired with the evader acting as a parent node; this topological relationship is fixed. Consequently, when assistance is required from another child node’s pursuers, the inability to change the topological relationship leads to an inability to leverage team effectiveness. In [17], the MPE game is decomposed into several MPE game subsystems. While the behavior within each subsystem does not influence the stability of the overall environment, there is an inherent lack of collaborative interaction between the subsystems themselves. These methods design value functions that take into account the state difference with respect to the goal or neighboring nodes. Although each distributed strategy is goal-oriented, agents within the same team lack communication and collaboration. Therefore, achieving coordinated control with team collaboration remains a challenge in many distributed control algorithms [19], [20]. Consequently, the complex interactions among multiple agents to enhance team cohesion merit in-depth research [21]. Consensus, which is extensively studied for multiagent system control using graph theory [22], is often achieved in the context of formation control by defining desired displacements [23]. In this work, while enhancing cohesion through target switching, the role of desired displacements is also considered.

Manuscript received 20 December 2023; revised 7 June 2024; accepted 20 July 2024. Date of publication 2 August 2024; date of current version 20 September 2024. This work was supported by the National Natural Science Foundation of China under Grant 62373302. (Corresponding author: Yan-Jun Liu.)

Zhiqiang Xu is with the Unmanned System Research Institute, Northwestern Polytechnical University, Xi’an 710072, China (e-mail: zhiqiangxu@mail.nwpu.edu.cn).

Dengxiu Yu and Zhen Wang are with the School of Artificial Intelligence, Optics and Electronics, Northwestern Polytechnical University, Xi’an 710072, China (e-mail: yudengxiu@nwpu.edu.cn; w-zhen@nwpu.edu.cn).

Yan-Jun Liu is with the College of Science, Liaoning University of Technology, Jinzhou 121001, China (e-mail: liuyanjun@live.com).

Digital Object Identifier 10.1109/JSYST.2024.3432796

In practical applications, several MPE game processes often involve inherent nonlinearities [9], [24]. Consequently, solving the nonlinear Hamilton–Jacobi–Isaacs (HJI) equations becomes necessary, for which analytical solutions are not readily available [25], [26]. To address this issue, various adaptive dynamic programming algorithms have been proposed [27], [28], [29], [30]. In addition, based on the principle of optimality, reinforcement learning algorithms employing the actor–critic (AC) architecture have been widely utilized for learning optimal policies online in MPE problems [29], [30]. However, in the context of multiagent systems, each agent is equipped with its own actor and critic networks [1]. The existence of multiple networks, while necessary for decentralized decision-making, inadvertently increases the computational burden [31]. To enhance computational efficiency, a single network adaptive critic (SNAC) structure has been proposed for solving control-constrained optimal control problems [32], which eliminates the need for an action network. This structure was later applied to single-pursuer single-evader games within the framework of fault-tolerant control [33]. Literature surveys reveal that existing studies have implemented the SNAC structure in the context of single-agent PE games and multiagent adversarial problems [33], [34]. However, no studies have been found to explore the potential application of this approach within MPE systems. Therefore, one of the objectives of this article is to explore the potential of this structure in the MPE system.

Another aspect worth mentioning is that, in engineering applications, the output amplitudes of actuators often exhibit saturated nonlinearities due to physical limitations [35]. Therefore, many studies also take into account the design of controllers considering the actuator saturation phenomenon [36], [37]. Ignoring the actuator saturation phenomenon not only increases the risk of performance degradation and instability in controlled systems but also results in a partial detachment of research from practicality. Furthermore, the stability and reliability of the systems are of utmost importance. Currently, mainstream approximate optimal control methods construct Lyapunov functions to prove the uniformly boundedness of nonlinear closed-loop systems. The stability analysis of multiagent systems based on graph-theoretic approaches is investigated in [38] and [34].

According to the aforementioned discussion, conventional MPE games present some interesting challenges in designing coordinated capture strategies among all pursuers. In this article, our objective is to design an algorithm that combines reinforcement learning and dynamic target graphs to approximate optimal control strategies for MPE games under actuator saturation conditions. The contributions of this article are summarized as follows.

- 1) A graph-theoretic approach is designed to represent the relationships among the agents, and based on this, a dynamic target graph algorithm is employed to enhance the coordination among the agents. Unlike the fixed communication graph in [17], [18], and [39], the weighted adjacency matrix of our graph is entirely dynamic, further improving the team cohesion among pursuers compared to [26].
- 2) In contrast to existing results [12], [26], which solved MPE games under linear systems, our work considers the nonlinear dynamic of the system and actuator constraints. Moreover, unlike [33], our method can obtain solutions for various scenarios with different numbers of pursuers

and evaders by solving the HJI equation, which implies that our algorithm faces a more complex environment.

- 3) In the proposed method, we utilize the value-based single network adaptive critic (V-SNAC) network structure, which reduces the computational burden compared to the AC network structure [39], [40]. We validate the effectiveness of our algorithm in two typical scenarios: multiple-pursuer-single-evader and multiple-pursuer-multiple-evader. In addition, a comparison of the computational burden with the AC structure [39] has been implemented, and the results indicate that the V-SNAC structure can reduce computational time by approximately 31%.

The rest of this article is organized as follows. Section II provides the preliminaries of the MPE problem, including the graph model and state equations. The solution strategies for the MPE game are proposed in Section III. The simulations and experiments for the two scenarios are presented in Section IV. Finally, Section V concludes this article.

II. PRELIMINARIES

A. Graph Theory

Consider an MPE scenario with N_p pursuers and N_e evaders, where the entire system consists of $N_p + N_e$ agents. Define the graph $\mathcal{G}_p = (\mathcal{V}_p, \mathcal{E}_p)$ to represent the communication topology of N_p pursuers. Set of nodes $\mathcal{V}_p = \{v_1^p, \dots, v_{N_p}^p\}$ and set of edges $\mathcal{E}_p \subseteq \mathcal{V}_p \times \mathcal{V}_p$. If and only pursuer m has access to the information of pursuer n , and we say agent n is a neighbor of agent m , i.e., $(v_m^p, v_n^p) \in \mathcal{E}_p$. Let $a_{m,n}$ be the communication weight of the graph $\mathcal{G}_p$, with $a_{m,n} = 1$ if $(v_m^p, v_n^p) \in \mathcal{E}_p$ and $a_{m,n} = 0$ otherwise. Let $\mathcal{A}_p = [a_{m,n}] \in \mathbb{R}^{N_p \times N_p}$ be the weighted adjacency matrix where $a_{m,m} = 0$. Define the in-degree of pursuer m as $d_m^p = \sum_{n=1}^{N_p} a_{m,n}$ and the in-degree matrix of the graph $\mathcal{D}_p = \text{diag}\{d_m^p\}$. Then, the Laplacian matrix can be defined as $\mathbf{L}_p = \mathcal{D}_p - \mathcal{A}_p$. Similarly, the graph $\mathcal{G}_e = (\mathcal{V}_e, \mathcal{E}_e)$ represents the interaction topology among N_e evaders with the nodes $\mathcal{V}_e = \{v_1^e, \dots, v_{N_e}^e\}$. The edge weights $b_{m,n}$ with $b_{m,n} = 1$ if $(v_m^e, v_n^e) \in \mathcal{E}_e$ and $b_{m,n} = 0$ otherwise. The in-degree of evader m as $d_m^e = \sum_{n=1}^{N_e} b_{m,n}$ and the in-degree matrix of the graph $\mathcal{D}_e = \text{diag}\{d_m^e\}$. Defined the matrix $\mathcal{A}_e = [b_{m,n}]$ as $\mathbf{L}_e = \mathcal{D}_e - \mathcal{A}_e$.

Define the bipartite graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represent the target topology among pursuers and evaders. Specifically, for $m \in \mathcal{V}_p$ and $n \in \mathcal{V}_e$, the edge weight $c_{m,n} = 1$ if pursuer m targets evader n and $c_{m,n} = 0$ otherwise. Similarly, $e_{n,m} = 1$ if evader n targets pursuer m for evasion and $e_{n,m} = 0$ otherwise. The in-degree of pursuer m in graph $\mathcal{G}$ is defined as $d_m^{\text{pe}} = \sum_{n=1}^{N_e} c_{m,n}$, and the in-degree of evader n is $d_n^{\text{ep}} = \sum_{m=1}^{N_p} e_{n,m}$. Define the pursuer-partition matrices $\mathcal{D}_{\text{pe}} = \text{diag}\{d_m^{\text{pe}}\}$ and $\mathcal{A}_{\text{pe}} = [c_{m,n}]$, and the evader-partition matrices $\mathcal{D}_{\text{ep}} = \text{diag}\{d_n^{\text{ep}}\}$ and $\mathcal{A}_{\text{ep}} = [e_{n,m}]$.

Assumption 1: For the MPE game problem in this article, it is assumed that agents on the same team can communicate and exchange information with each other. Each pursuer has access to the state information of their targeted evader, and conversely, each evader can obtain information about the pursuers who are targeting them as their objectives. Graphs $\mathcal{G}_p$ and $\mathcal{G}_e$ are connected graphs.

B. Problem Formulation

Consider a system comprising N_p pursuers, each characterized by the following dynamics:

$$\dot{x}_j^p = f(x_j^p) + g(x_j^p) u_j^p \quad (1)$$

where $j = 1, \dots, N_p$, $x_j^p \in \mathbb{R}^n$, is the system state of the j th pursuer, $u_j^p \in \mathbb{R}^m$ denotes the control input of the j th pursuer.

Consider also a system with N_e evaders, each described by

$$\dot{x}_i^e = f(x_i^e) + g(x_i^e) u_i^e \quad (2)$$

where $i = 1, \dots, N_e$, $x_i^e \in \mathbb{R}^n$, is the system state of the i th evader, $u_i^e \in \mathbb{R}^m$ denotes the control input of the i th evader. u_j^p and u_i^e are both subjected to saturation constraint $\bar{u}_p$ and $\bar{u}_e$, respectively, i.e., $|u_j^p| \leq \bar{u}_p$ and $|u_i^e| \leq \bar{u}_e$. $\bar{u}_p$ and $\bar{u}_e$ are positive constant. $f(\cdot) \in \mathbb{R}^n$ is the continuous nonlinear function and $g(\cdot) \in \mathbb{R}^{n \times m}$ is the continuous control gain function. In this article, we assumed that $f(\cdot), g(\cdot)$ are Lipchitz functions on a compact set Ω_1 .

For each evader, define the state error variable between the pursuers and their target evader i as

$$\tilde{x}_i^{\text{pe}} = \sum_{j=1}^{N_p} (c_{j,i}(x_j^p - x_i^e + r_{j,i}^{\text{pe}})) \quad (3)$$

where $r_{j,i}^{\text{pe}}$ is the expected displacement between the pursuer j and the evader i , and it can be a zero vector if necessary.

Using (1) and (2), it can be inferred that

$$\dot{\tilde{x}}_i^{\text{pe}} = \sum_{j=1}^{N_p} (c_{j,i}(f(x_j^p) - f(x_i^e) + g(x_j^p) u_j^p - g(x_i^e) u_i^e)). \quad (4)$$

Remark 1: In the MPE game framework considered in this article, we assume that the number of pursuers is greater than or equal to the number of evaders, i.e., $N_p \geq N_e$. If the number of evaders is greater, there will inevitably be some evaders escaping without facing any challenge, which is not within the scope of our analysis. Conversely, if the number of pursuers is greater, there can be cases where several pursuers simultaneously chase a single evader. The pursuers may aim to intercept the target evader by achieving surrounding formation control. The target allocation method, which will be discussed later in this article, contributes to better team cohesion among pursuers.

When considering the PE game involving multiple pursuers and evaders, the augmented system states are defined as follows:

$$X^p = \begin{bmatrix} x_1^p \\ x_2^p \\ \vdots \\ x_{N_p}^p \end{bmatrix}, \quad X^e = \begin{bmatrix} x_1^e \\ x_2^e \\ \vdots \\ x_{N_e}^e \end{bmatrix} \quad (5)$$

then the multiagent PE system based on (1) and (2) can be described as

$$\begin{aligned} \dot{X}^p &= F(X^p) + \sum_{j=1}^{N_p} G_j(X^p) u_j^p \\ \dot{X}^e &= F(X^e) + \sum_{i=1}^{N_e} G_i(X^e) u_i^e \end{aligned} \quad (6)$$

where $F(X^p) = [f^T(x_1^p), \dots, f^T(x_{N_p}^p)]^T$, $G_j(X^p) = [B_1, \dots, B_j, \dots, B_{N_p}]^T \otimes [g^T(x_1^p), \dots, g^T(x_{N_p}^p)]^T$ with $B_m = 0_n$ for $m = 1, \dots, N_p$, $m \neq j$ and $B_m = I_n$ if $m = j$. In

Algorithm 1: Dynamic Target Graph Algorithm.

Input: System states of pursuers $\{x_j^p\}$ and evaders $\{x_i^e\}$. Specify the swap threshold $\varrho \geq 0$

Calculate the distance $\gamma_{j,i}$ between all pursuers and evaders.

For evader $e_i, i \leq N_e$ and pursuers $p_j, j \leq N_p$

 If $(\gamma_{j,i} + \gamma_{j',i'}) - (\gamma_{j',i} + \gamma_{j,i'}) > \varrho$

 Swap their edge weights that exchanging $c_{j,i}$ with $c_{j',i}$ and $c_{j,i'}$ with $c_{j',i'}$. Update $e_{j,i}$ accordingly.

until comparison of other distances γ is completed.

 Update the weighted adjacency matrix $\mathcal{A}_{pe}$ and $\mathcal{A}_{ep}$.

Output: $\{\mathcal{A}_{pe}, \mathcal{A}_{ep}\}$.

which $\otimes$ stands for Kronecker product. Similarly, $F(X^e) = [f^T(x_1^e), \dots, f^T(x_{N_e}^e)]^T$ and $G_i(X^e) = [D_1, \dots, D_i, \dots, D_{N_e}]^T \otimes [g^T(x_1^e), \dots, g^T(x_{N_e}^e)]^T$ with $D_m = 0_n$ for $m = 1, \dots, N_e$, $m \neq i$ and $D_m = I_n$ if $m = i$.

In the following, for simplicity, we will write $\{f_j^p, f_i^e\}$ and $\{g_j^p, g_i^e\}$ to replace $\{f(x_j^p), f(x_i^e)\}$ and $\{g(x_j^p), g(x_i^e)\}$, respectively.

III. OBJECTIVE DESIGN AND SOLUTIONS

A. Target Allocation and Objective Design

Define $\gamma_{j,i} > 0$ as the weighted 2-norm of the state difference between a specific pursuer j and another evader i , given by $\gamma_{j,i} = \|\nu \odot (x_j^p - x_i^e + r_{j,i}^{\text{pe}})\|$, where $\nu \in \mathbb{R}^n$ is the element weight and $\odot$ stands for Hadamard product. Both velocity and position information are contained in the state. If there exists another pair of pursuer j' and its target evader i' , where $c_{j',i} = 0$ or $c_{j,i'} = 0$, it implies that pursuer j' and evader i , as well as pursuer j and evader i' , do not form a target relationship. However, by exchanging their respective evader targets i and i' , pursuers j and j' can achieve their objectives more easily. Consequently, the current weighted adjacency matrices $\mathcal{A}_{pe}$ and $\mathcal{A}_{ep}$ will be adjusted according to Algorithm 1.

Remark 2: The swap threshold ϱ is crucial for ensuring significant benefits from target switches by pursuers. Increasing ϱ raises the switching barrier, potentially reducing the frequency or possibility of switches. Decreasing ϱ makes switching easier and more frequent, but excessive switching for trivial gains is impractical in engineering. The simulation section will carefully determine an appropriate ϱ value.

In order to quantify and measure the overall team cohesion of the pursuers, we define the team state error of pursuers with target topology obtained by Algorithm 1 as

$$E_{\text{Team}} = \sum_{i=1}^{N_e} \sum_{j=1}^{N_p} (c_{j,i} \|\nu \odot (x_j^p - x_i^e + r_{j,i}^{\text{pe}})\|). \quad (7)$$

The team state error can reflect the overall state error within the team rather than the error of individual agents. The smaller the team state error, the stronger the team cohesion is considered to be. Subsequently, we analyze the enhancement of cohesion from a mathematical perspective.

Theorem 1: Consider two arbitrary scenarios A and B, each characterized by identical MPE game dynamics described in (6). Scenario A uses algorithm 2 to dynamically allocate targets, and scenario B uses the initial fixed target topology. The team state

errors of scenarios A and B are defined as E_{Team} and E_{Team}^O , respectively. Then, $E_{\text{Team}} \leq E_{\text{Team}}^O$ holds.

Proof: Based on Algorithm 1, If the condition $(\gamma_{j,i} + \gamma_{j',i'}) - (\gamma_{j',i} + \gamma_{j,i'}) > \varrho$ is not triggered, due to the same initial target topology, the two methods have the same adjacency matrix from beginning to end, and subsequently $E_{\text{Team}} = E_{\text{Team}}^O$.

If at a certain time step t , two pursuers exchange their target evaders, it can be described as $c_{j,i} = 0, c_{j',i'} = 0, c_{j,i} = 1, c_{j,i'} = 1$ to the state where $c_{j,i} = 1, c_{j',i'} = 1, c_{j,i} = 0, c_{j,i'} = 0$. Let $\tilde{x}_{\text{uc}}$ and $\tilde{x}_{\text{uc}}^O$ be defined as the sums of the unchanged target topology part of the state errors for E_{Team} and E_{Team}^O , respectively. Then, E_{Team} and E_{Team}^O can be reformulated as

$$\begin{aligned} E_{\text{Team}} &= \tilde{x}_{\text{uc}} + \left\| \nu \odot (x_{j'}^p - x_i^e + r_{j,i}^{\text{pe}}) \right\| \\ &\quad + \left\| \nu \odot (x_j^p - x_{i'}^e + r_{j,i}^{\text{pe}}) \right\| \\ E_{\text{Team}}^O &= \tilde{x}_{\text{uc}}^O + \left\| \nu \odot (x_j^p - x_i^e + r_{j,i}^{\text{pe}}) \right\| \\ &\quad + \left\| \nu \odot (x_{j'}^p - x_{i'}^e + r_{j,i}^{\text{pe}}) \right\|. \end{aligned} \quad (8)$$

Based on (8), we calculate the difference between the team error states E_{Team} and E_{Team}^O at time step t , i.e.,

$$\begin{aligned} E_{\text{Team}} - E_{\text{Team}}^O &= \left\| \nu \odot (x_{j'}^p - x_i^e + r_{j,i}^{\text{pe}}) \right\| \\ &\quad + \left\| \nu \odot (x_j^p - x_{i'}^e + r_{j,i}^{\text{pe}}) \right\| - \left\| \nu \odot (x_j^p - x_i^e + r_{j,i}^{\text{pe}}) \right\| \\ &\quad - \left\| \nu \odot (x_{j'}^p - x_{i'}^e + r_{j,i}^{\text{pe}}) \right\|. \end{aligned} \quad (9)$$

Since

$$\begin{aligned} &(\gamma_{j,i} + \gamma_{j',i'}) - (\gamma_{j',i} + \gamma_{j,i'}) \\ &= \left\| \nu \odot (x_j^p - x_i^e + r_{j,i}^{\text{pe}}) \right\| + \left\| \nu \odot (x_{j'}^p - x_{i'}^e + r_{j,i}^{\text{pe}}) \right\| \\ &\quad - \left\| \nu \odot (x_{j'}^p - x_i^e + r_{j,i}^{\text{pe}}) \right\| - \left\| \nu \odot (x_j^p - x_{i'}^e + r_{j,i}^{\text{pe}}) \right\| > \varrho. \end{aligned} \quad (10)$$

It can be deduced that

$$E_{\text{Team}} - E_{\text{Team}}^O \leq -\varrho < 0. \quad (11)$$

Since the previous values of $c_{j,i}, c_{j',i'}, c_{j',i}$, and $c_{j,i'}$ are actually arbitrarily chosen, therefore, by the same token, any subsequent exchange will satisfy $E_{\text{Team}} \leq E_{\text{Team}}^O$, thereby mathematically demonstrating the enhancement of cohesion. ■

Based on Algorithm 1 and Theorem 1, dynamic target assignment prioritizes team cohesion over individual optimization by reducing the pursuer team's overall cost. To quantify the outcome of the MPE game, for each target evader, the cost function J_i is defined as

$$J_i(x_0, u_j^p, u_i^e) = \int_0^\infty (P_i(x) + U(u_j^p) - W(u_i^e)) dt \quad (12)$$

where $P_i(x)$ is defined as

$$P_i(x) = (\tilde{x}_i^{\text{pe}})^T Q \tilde{x}_i^{\text{pe}} \quad (13)$$

and [32]

$$\begin{cases} U(u_j^p) = \sum_{j=1}^{N_p} \left(2c_{i,j} \int_0^{u_j^p} (\bar{u}_p \tanh^{-1} \frac{\tau}{\bar{u}_p})^T R_1 d\tau \right) \\ W(u_i^e) = 2d_i^{\text{ep}} \int_0^{u_i^e} (\bar{u}_e \tanh^{-1} \frac{\tau}{\bar{u}_e})^T R_2 d\tau \end{cases} \quad (14)$$

with positive-definite matrices $Q \in \mathbb{R}^{n \times n}$ and $R_1, R_2 \in \mathbb{R}^{m \times m}$. Taking into the factor of actuator saturation, non-quadratic functions $U(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}$ and $W(\cdot) : \mathbb{R}^m \rightarrow \mathbb{R}$ are

designed to constrain the control input u_j^p and u_i^e within $\bar{u}_p$ and $\bar{u}_e$, respectively.

Definition 1: A series of control policies $u_j^p = \{u_1^p, \dots, u_{N_p}^p\}$ and $u_i^e = \{u_1^e, \dots, u_{N_e}^e\}$ are defined to be admissible with respect to (12) over $\Omega_2 \in \mathbb{R}^n$, if $\{u_j^p, u_i^e\}$ are continuous over Ω_2 and the control pair $\{u_j^p, u_i^e\}$ stabilizes (6) over Ω_2 , and (12) is finite.

According to the control objective, pursuers aim to minimize function $J_i(x_0, u_j^p, u_i^e)$, while the evaders aim to maximize the performance function $J_i(x_0, u_j^p, u_i^e)$ by designing their respective control input u_j^p and u_i^e . Upon achieving equilibrium conditions in the PE game, as per the Nash–Pontryagin minimax principle, this state is referred to as a saddle point equilibrium (SPE). There is no motivation for any agents to deviate from the equilibrium. In the following, the definition of SPE is presented [41].

Definition 2: Consider the MPE game in this article. Every pair of control strategies (u_j^{p*}, u_i^{e*}) is said to be the SPE solution if

$$J_i(x(0), u_j^{p*}, u_i^e) \leq J_i(x(0), u_j^{p*}, u_i^{e*}) \leq J_i(x(0), u_j^p, u_i^{e*}). \quad (15)$$

Define the value functions associated with admissible control policy as

$$V_i(x) = \int_t^\infty (P_i(x) + U(u_j^p) - W(u_i^e)) d\tau. \quad (16)$$

The optimal value functions $V_i^*(x)$ is defined as

$$V_i^*(x) = \min_{\{u_j^p, u_i^e\} \in \Omega_2} \left(\int_t^\infty (P_i(x) + U(u_j^p) - W(u_i^e)) d\tau \right). \quad (17)$$

Based on the fact that the value functions (16) are continuously differentiable, taking its differentiation and obtain the Hamiltonian equation

$$\begin{aligned} H_i(x, u_j^p, u_i^e, \nabla V) &= P_i(x) + U(u_j^p) - W(u_i^e) \\ &\quad + \nabla^T V_i(x) \left(\sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right) \end{aligned} \quad (18)$$

where $\nabla V_i(x) \triangleq \frac{\partial V_i(x)}{\partial \tilde{x}_i^{\text{pe}}}$, $\nabla V_i(x) \in \mathbb{R}^n$, and $\mathcal{F}_{j,i}(x) \triangleq f_j^p - f_i^e$.

Taking the first-order derivative of H_i with respect to u_j^p and u_i^e , respectively, we can obtain that

$$\begin{cases} u_j^p = -\bar{u}_p \tanh \left(\frac{1}{2\bar{u}_p} R_1^{-1} (g_j^p)^T \nabla V_i(x) \right) \\ u_i^e = -\bar{u}_e \tanh \left(\frac{1}{2\bar{u}_e} R_2^{-1} (g_i^e)^T \nabla V_i(x) \right) \end{cases} \quad (19)$$

According to [42], we have

$$\begin{aligned} H_i(x, u_j^{p*}, u_i^{e*}, \nabla V_i^*) &= \max_{u_i^e} \min_{u_j^p} H_i(x, u_i^e, u_j^p, \nabla V_i^*) \\ &= \min_{u_j^p} \max_{u_i^e} H_i(x, u_i^e, u_j^p, \nabla V_i^*) = 0. \end{aligned} \quad (20)$$

For the brevity, we will write $J_i^*(x_0)$, H_i^* to denote $J_i^*(x_0, u_j^{p*}, u_i^{e*})$, $H_i(x, u_j^{p*}, u_i^{e*}, \nabla V_i^*)$ subsequently.

B. PI Algorithm for MPE Game

In this section, the HJI equation will be introduced, which is used to derive the optimal control strategy $\{u_j^*, u_i^{e*}\}$ and the value functions $V_i^*(x)$. The concept of admissible control is defined in Definition 1. Then, the subsequent best-response HJI equation, serving as the Bellman equation for the system, is defined as

$$H_i^* = P_i(x) + U(u_j^{p*}) - W(u_i^{e*}) + \nabla^T V_i(x) \left(\sum_{j=1}^{N_p} (c_{j,i}(\mathcal{F}_{j,i}(x) + g_j^p u_j^{p*} - g_i^e u_i^{e*})) \right) = 0 \quad (21)$$

where u_j^{p*} and u_i^{e*} represent the optimal control strategies, which are given by

$$\begin{cases} u_j^{p*} = \arg \min_{u_j^p \in \Omega_2} H_i(x, u_j^p, u_i^e, \nabla V_i) \\ \quad = -\bar{u}_p \tanh \left(\frac{1}{2\bar{u}_p} R_1^{-1} (g_j^p)^T \nabla V_i^*(x) \right) \\ u_i^{e*} = \arg \max_{u_i^e \in \Omega_2} H_i(x, u_j^p, u_i^e, \nabla V_i) \\ \quad = -\bar{u}_e \tanh \left(\frac{1}{2\bar{u}_e} R_2^{-1} (g_i^e)^T \nabla V_i^*(x) \right) \end{cases} \quad (22)$$

Substituting (22) into (21), the approximate HJI function is expressed in the following form:

$$\begin{aligned} H_i^* &= P_i(x) \\ &+ \sum_{j=1}^{N_p} \left(2c_{j,i} \int_0^{-\bar{u}_p \tanh(\rho_p(V_i^*))} \left(\bar{u}_p \tanh^{-1} \frac{\tau}{\bar{u}_p} \right)^T R_1 d\tau \right) \\ &- 2d_i^{ep} \int_0^{-\bar{u}_e \tanh(\rho_e(V_i^*))} \left(\bar{u}_e \tanh^{-1} \frac{\tau}{\bar{u}_e} \right)^T R_2 d\tau \\ &+ \nabla^T V_i^*(x) \left(\sum_{j=1}^{N_p} (c_{j,i}(\mathcal{F}_{j,i}(x) + g_j^p u_j^{p*} - g_i^e u_i^{e*})) \right) = 0 \end{aligned} \quad (23)$$

where $\rho_p(V_i^*), \rho_e(V_i^*) \in \mathbb{R}^m$ are defined as

$$\begin{aligned} \rho_p(V_i^*) &\triangleq \frac{1}{2\bar{u}_p} R_1^{-1} (g_j^p)^T \nabla V_i^*(x) \\ \rho_e(V_i^*) &\triangleq \frac{1}{2\bar{u}_e} R_2^{-1} (g_i^e)^T \nabla V_i^*(x). \end{aligned}$$

Then, (23) can be reformulated as

$$\begin{aligned} H_i^* &= P_i(x) \\ &+ \sum_{j=1}^{N_p} (c_{j,i} \bar{u}_p \tanh^T(\rho_p(V_i^*)) (g_j^p)^T \nabla V_i^*(x) \\ &+ c_{j,i} \bar{u}_p^2 \bar{R}_1 \ln(\bar{\mathbf{I}} - \tanh^2(\rho_p(V_i^*))) \\ &- d_i^{ep} (\bar{u}_e \tanh^T(\rho_e(V_i^*)) (g_i^e)^T \nabla V_i^*(x) \\ &+ \bar{u}_e^2 \bar{R}_2 \ln(\bar{\mathbf{I}} - \tanh^2(\rho_e(V_i^*))) \\ &- \nabla^T V_i^*(x) \sum_{j=1}^{N_p} (c_{j,i} g_j^p \bar{u}_p \tanh(\rho_p(V_i^*))) \\ &+ \nabla^T V_i^*(x) d_i^{ep} g_i^e \bar{u}_e \tanh(\rho_e(V_i^*)) \\ &+ \nabla^T V_i^*(x) \left(\sum_{j=1}^{N_p} (c_{j,i} \mathcal{F}_{j,i}(x)) \right) = 0 \end{aligned} \quad (24)$$

Algorithm 2: Policy Iteration Algorithm.

Input: System initial states $\{x_j^p, x_i^e\}$ and their corresponding initial admissible control policy $\{\mu_{j,0}^p, \mu_{i,0}^e\}$.

Initialize the error threshold ξ . Set $V_i^0 = 0$, and let $s = 0$.

repeat

Based on the control pair $\{\mu_{j,s}^p, \mu_{i,s}^e\}$, solve the equation (26) to obtain the value functions V_i^{s+1}

Update the control policy $\mu_{j,s}^p$ and $\mu_{i,s}^e$ simultaneously by

$$\begin{aligned} \mu_{j,s}^p &= -\bar{u}_p \tanh \left(\frac{1}{2\bar{u}_p} R_1^{-1} (g_j^p)^T \nabla V_i^{s+1}(x) \right) \\ \mu_{i,s}^e &= -\bar{u}_e \tanh \left(\frac{1}{2\bar{u}_e} R_2^{-1} (g_i^e)^T \nabla V_i^{s+1}(x) \right). \end{aligned} \quad (25)$$

Update the weighted adjacency matrix $\mathcal{A}_{pe}$ and $\mathcal{A}_{ep}$ simultaneously by using Algorithm 1.

Set $s = s + 1$.

until $|V_i^s - V_i^{s-1}| < \xi$

Output: $\{\mu_{j,s}^p, \mu_{i,s}^e\}$.

where $\bar{\mathbf{I}} \triangleq [1 \ 1 \ \dots \ 1]^T \in \mathbb{R}^m$, $\bar{R}_1 \triangleq [r_{11} \ r_{12} \ \dots \ r_{1m}]$, and $\bar{R}_2 \triangleq [r_{21} \ r_{22} \ \dots \ r_{2m}]$.

Remark 3: In this article, the control strategy for each agent is distributed. However, in Algorithm 2, each agent's solution of the HJI equation requires knowledge of the dynamics of the associated agent, which is implied in $\nabla V_i^{s+1}(x)$ in (25). This restricts the application of the results. Therefore, in the next section, we will improve Algorithm 2 to achieve distributed control.

Thus, the approximate HJI function can be simplified as

$$\begin{aligned} H_i^* &= P_i(x) \\ &+ \sum_{j=1}^{N_p} \left(c_{j,i} \bar{u}_p^2 \bar{R}_1 \ln(\bar{\mathbf{I}} - \tanh^2(\rho_p(V_i^*))) \right) \\ &- d_i^{ep} \bar{u}_e^2 \bar{R}_2 \ln(\bar{\mathbf{I}} - \tanh^2(\rho_e(V_i^*))) \\ &+ \nabla^T V_i^*(x) \left(\sum_{j=1}^{N_p} (c_{j,i} \mathcal{F}_{j,i}(x)) \right) = 0. \end{aligned} \quad (26)$$

In this article, we denote the functions $V_i^s(x)$ and $\{\mu_{1,s}^p, \mu_{1,s}^e, \dots, \mu_{N_p,s}^p, \mu_{N_e,s}^e\}$ (for $s = 1, 2, \dots$) as approximations of the optimal values $V_i^*(x)$ and $\{u_j^*, u_i^{e*}\}$ at each iteration step s . Then, to solve the HJI approximate equations, the on-policy PI algorithm is put forward in Algorithm 2.

In the realm of PE games involving nonlinear systems, Yuan et al. [33] conducted pertinent research. However, the property for MPE games has not been sufficiently explored. In the following, we will demonstrate the convergence of the on-policy PI algorithm for MPE games. The analysis is based on the work of [43].

Theorem 2: Consider the system of MPE game described by (6) with admissible control pair $\{\mu_{j,s}^p, \mu_{i,s}^e\}$ obtained by (25) in Algorithm 2. If the system model parameters $F(X^p)$, $F(X^e)$ and $G_i(X^p)$, $G_i(X^e)$ are known, then the iterative value functions will converge to $V_i^*(x)$, as $s \rightarrow \infty$.

Proof: Based on (26), we consider a Banach space [44] $\mathbb{V} \subset \{V_i(x) \mid V_i(x) : \Omega \rightarrow \mathbb{R}, V_i(0) = 0\}$ with a norm $\|\cdot\|_\Omega$. Then, define the mapping $\mathcal{H}_i(V_i(x)) : \underbrace{\mathbb{V} \times \mathbb{V} \cdots \times \mathbb{V}}_{N_e} \rightarrow \mathbb{V}$ as

$$\begin{aligned} \mathcal{H}_i(V_i(x)) &\triangleq P_i(x) \\ &+ \sum_{j=1}^{N_p} \left(c_{j,i} \bar{u}_p^2 \bar{R}_1 \ln \left(\vec{\mathbf{1}} - \tanh^2(\rho_p(V_i)) \right) \right) \\ &- d_i^{\text{ep}} \bar{u}_e^2 \bar{R}_2 \ln \left(\vec{\mathbf{1}} - \tanh^2(\rho_e(V_i)) \right) \\ &+ \nabla^T V_i(x) \left(\sum_{j=1}^{N_p} (c_{j,i} \mathcal{F}_{j,i}(x)) \right) = 0. \end{aligned} \quad (27)$$

Define mapping $\mathcal{T}_i : \mathbb{V} \rightarrow \mathbb{V}$

$$\mathcal{T}_i(V_i(x)) \triangleq V_i(x) - (\mathcal{H}'_{iV_i})^{-1} \mathcal{H}_i(V_i(x)) \quad (28)$$

where $\mathcal{H}'_{iV_i}$ denotes the Fréchet derivative of $\mathcal{H}_i$ taken with respect to V_i . Here, we introduce the definition of Gâteaux derivative to compute the Fréchet derivative indirectly.

Definition 3: Let $\mathcal{H} : \mathbb{U} \subseteq \mathbb{X} \rightarrow \mathbb{Y}$ be a given map with $\mathbb{X}$ and $\mathbb{Y}$ Banach spaces, where $\mathbb{U}(V)$ denotes a neighborhood of V . The map $\mathcal{H}$ is Gâteaux differentiable at V if and only there exists a bounded linear operator $\mathcal{L} : \mathbb{X} \rightarrow \mathbb{Y}$ such that [25], [45]

$$\mathcal{H}(V + tK) - \mathcal{H}(V) = t\mathcal{L}(K) + o(s), s \rightarrow 0$$

for all K with $\|K\|_\Omega = 1$ and all real numbers t in some neighborhood of zero, where $\lim_{s \rightarrow 0} (o(s)/s) = 0$. $\mathcal{L}$ is called the Gâteaux derivative of $\mathcal{H}$ at V . Then, the Gâteaux derivative at V is defined as

$$\mathcal{L}(K) = \lim_{t \rightarrow 0} \frac{\mathcal{H}(V + tK) - \mathcal{H}(V)}{t}. \quad (29)$$

To compute the Fréchet derivative, we introduce the following Lemma 1.

Lemma 1 (see [46]): If the Gâteaux derivative $\mathcal{L}$ exists in some neighborhood of V and $\mathcal{L}$ is continuous at V , then $\mathcal{L} = \mathcal{H}'(V)$ is also an Fréchet derivative at V .

Then, we compute the Fréchet derivative in the following Lemma 2 by (29).

Lemma 2: Consider the mapping $\mathcal{H}_i$ defined in (27), $\forall V_i \in \mathbb{V}$, the Gâteaux differential of $\mathcal{H}_i$ at V_i is

$$\begin{aligned} \mathcal{H}'_{iV_i} K &= \mathcal{L}_{iV_i^{s+1}}(K) \\ &= \nabla^T K \left(\sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right) \\ &= 0. \end{aligned} \quad (30)$$

Proof: First, according to the definition of $\mathcal{H}_i$ in (27), $\forall V_i \in \mathbb{V}$, one has

$$\begin{aligned} \mathcal{H}_i(V_i(x) + tK) - \mathcal{H}_i(V_i(x)) \\ &= \sum_{j=1}^{N_p} \left(c_{j,i} \bar{u}_p^2 \bar{R}_1 \left(\ln \left(\vec{\mathbf{1}} - \tanh^2(\rho_p(V_i + tK)) \right) \right. \right. \\ &\quad \left. \left. - \ln \left(\vec{\mathbf{1}} - \tanh^2(\rho_p(V_i)) \right) \right) \right) \\ &\quad - d_i^{\text{ep}} \bar{u}_e^2 \bar{R}_2 \left(\ln \left(\vec{\mathbf{1}} - \tanh^2(\rho_e(V_i + tK)) \right) \right. \\ &\quad \left. - \ln \left(\vec{\mathbf{1}} - \tanh^2(\rho_e(V_i)) \right) \right) \end{aligned}$$

$$\begin{aligned} &- \ln \left(\vec{\mathbf{1}} - \tanh^2(\rho_e(V_i)) \right) \\ &+ t \nabla^T K(x) \left(\sum_{j=1}^{N_p} (c_{j,i} \mathcal{F}_{j,i}(x)) \right). \end{aligned}$$

Therefore, the Gâteaux differential at V_i is

$$\begin{aligned} \mathcal{L}_{iV_i}(K) &= \lim_{t \rightarrow 0} \frac{\mathcal{H}_i(V_i + tK) - \mathcal{H}_i(V_i)}{t} \\ &= \nabla^T K \left(\sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right). \end{aligned} \quad (31)$$

Then, we will demonstrate the continuity of $\mathcal{L}_{iV_i}$ on V_i . $\forall K_0 \in \mathbb{V}$, we have

$$\begin{aligned} &\|\mathcal{L}_{iV_i}(K) - \mathcal{L}_{iV_i}(K_0)\|_\Omega \\ &= \left\| \nabla^T (K - K_0) \left(\sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right) \right\|_\Omega \\ &\leq \left\| \sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right\|_\Omega \|\nabla(K - K_0)\|_\Omega \\ &\leq \left\| \sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right\|_\Omega \varphi \|K - K_0\|_\Omega \\ &\triangleq \varpi \|K - K_0\|_\Omega \end{aligned}$$

where $\varphi > 0$ and ϖ is defined as

$$\varpi \triangleq \varphi \left\| \sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right\|_\Omega.$$

Therefore, $\varepsilon > 0$, $\exists \eta = \frac{\varepsilon}{\varpi}$ such that

$$\|\mathcal{L}_{iV_i}(K) - \mathcal{L}_{iV_i}(K_0)\|_\Omega \leq \varpi \|K - K_0\|_\Omega < \varepsilon.$$

Since $\varpi \|K - K_0\|_\Omega < \eta$, which implies $\mathcal{L}_{iV_i}$ is continuous at V_i . According to Lemma 1, $\mathcal{H}'_{iV_i^{s+1}} K = \mathcal{L}_{iV_i^{s+1}}(K)$ corresponds to the Fréchet differential as depicted in (29), while $\mathcal{H}'_{iV_i^{s+1}} = \mathcal{L}_{iV_i^{s+1}}$ represents the Fréchet derivative. ■

In accordance with the conclusion of the Lemma 2, we can prove that the policy iteration algorithm is mathematically equivalent to the quasi-Newton's iteration in a Banach space $\mathbb{V}$.

Theorem 3: Consider the mapping $\mathcal{T}$ defined in (28), then the iterative algorithm between (26) and (25) is equivalent to the following quasi-Newton's iteration:

$$\begin{aligned} V_i^{s+1}(x) &= \mathcal{T}(V_i^s(x)) \\ &= V_i^s - (\mathcal{H}'_{iV_i})^{-1} \mathcal{H}_i(V_i(x)), s = 0, 1, 2, \dots \end{aligned} \quad (32)$$

Proof: Considering (30) and Lemma 2, one has

$$\begin{aligned} \mathcal{H}'_{iV_i^{s+1}} V_i^{s+1}(x) \\ &= \nabla^T V_i^{s+1}(x) \left(\sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right) \end{aligned} \quad (33)$$

and

$$\begin{aligned} & \mathcal{H}'_{V_i^s} V_i^s(x) \\ &= \nabla^T V_i^s(x) \left(\sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right). \end{aligned} \quad (34)$$

Combining (25) and (27), one has

$$\begin{aligned} \mathcal{H}(V_i^s(x)) &= P_i(x) + U(u_j^p) - W(u_i^e) \\ &+ \nabla^T V_i^s(x) \left(\sum_{j=1}^{N_p} (c_{j,i} (\mathcal{F}_{j,i}(x) + g_j^p u_j^p - g_i^e u_i^e)) \right). \end{aligned} \quad (35)$$

Subtracting (34) from (35), we have

$$\begin{aligned} \mathcal{H}'_{V_{i^{s+1}}} V_{i^{s+1}}(x) - \mathcal{H}(V_i^s(x)) \\ = -(P_i(x) + U(u_j^p) - W(u_i^e)). \end{aligned} \quad (36)$$

Considering (24) and (33), (36), we obtain

$$\mathcal{H}'_{V_{i^{s+1}}} V_{i^{s+1}}(x) = \mathcal{H}'_{V_{i^{s+1}}} V_i^s(x) - \mathcal{H}(V_i^s(x))$$

which is accordant with (32). ■

Since the proposed on-policy policy iteration algorithm between the iteration is equivalent to the quasi-Newton's iteration, we can deduce that the value functions $V_i^s(x)$ will eventually converge to the optimal value $V_i^*(x)$ as $s \rightarrow \infty$. As a result, $\mu_{j,s}^p, \mu_{i,s}^e$ will tend to follow the optimal value according to (22), simultaneously. Therefore, Theorem 2 holds. ■

C. Off-Policy RL Algorithm Design

In this section, an off-policy RL algorithm based on the foundation of Algorithm 2 is proposed. Note that Algorithm 2 requires the knowledge of value functions $V_i(x)$ to solve the approximate HJI equation. Thus, we leverage the approximating characteristics of neural networks to address this issue. In the case of multiple agents, the application of multiple networks introduces computational burden. Therefore, we adopt the V-SNAC structure. In this structure, for each evader, only one critic network is utilized to captures the mapping between states and the value functions $V_i(x)$. In MPE game, all the agents will continuously and simultaneously update the network weights. Therefore, the value functions $V_i(x)$ can be approximated by critic network as

$$V_i^*(x) = W_i^T \psi_i(x) + \delta_i(x) \quad (37)$$

where $W_i = [W_{i,1}, W_{i,2}, \dots, W_{i,h}]^T \in \mathbb{R}^h$ are the ideal weight vectors, h stands for the number of the hidden neurons, $\psi_i(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^h$ denotes the nonlinear activation functions, and $\delta_i(x) \in \mathbb{R}$ represents the approximation error of the neural networks.

And the value solution $V_i^*(x)$ approximated by critic network at step s are reconstructed as

$$\hat{V}_i^s(x) = \hat{W}_{i,s}^T \psi_i(x) \quad (38)$$

where $\hat{V}_i^s(x)$ represents the estimate of the optimal value $V_i^*(x)$ and $\hat{W}_{i,s} \in \mathbb{R}^h$ represents the estimated weight vector at step s .

The derivatives of $\hat{V}_i^s(x)$ with respect to $\tilde{x}_i^{\text{pe}}$ are represented as

$$\nabla \hat{V}_i(x) = \nabla \psi_i^T(x) \hat{W}_{i,s} \quad (39)$$

where $\nabla \psi_i^T(x) \in \mathbb{R}^{n \times h}$ are bounded gradients of the activation functions.

To circumvent the need for system dynamics, the neural network is employed to approximate the value functions. Substituting (39) into (22) yields

$$\begin{cases} u_j^{p*} = -\bar{u}_p \tanh\left(\frac{1}{2\bar{u}_p} R_1^{-1} (g_j^p)^T \nabla \psi_i^T(x) \hat{W}_{i,s}\right) \\ u_i^{e*} = -\bar{u}_e \tanh\left(\frac{1}{2\bar{u}_e} R_2^{-1} (g_i^e)^T \nabla \psi_i^T(x) \hat{W}_{i,s}\right) \end{cases} \quad (40)$$

and substituting (40) into (26) yields

$$\begin{aligned} H_i^* &= P_i(x) \\ &+ \sum_{j=1}^{N_p} \left(c_{j,i} \bar{u}_p^2 \bar{R}_1 \ln \left(\bar{\mathbf{I}} - \tanh^2(\rho_p(\nabla \psi_i^T(x) \hat{W}_{i,s})) \right) \right) \\ &- d_i^{\text{ep}} \bar{u}_e^2 \bar{R}_2 \ln \left(\bar{\mathbf{I}} - \tanh^2(\rho_e(\nabla \psi_i^T(x) \hat{W}_{i,s})) \right) \\ &+ \hat{W}_{i,s}^T \nabla \psi_i(x) \left(\sum_{j=1}^{N_p} (c_{j,i} \mathcal{F}_{j,i}(x)) \right) = 0. \end{aligned} \quad (41)$$

Equation (41) is employed as the off-policy Bellman equation. From (26) and (40), derive the definition of the residual error e_i^s at step s

$$\begin{aligned} e_i^s &\triangleq (\tilde{x}_i^{\text{pe}})^T Q \tilde{x}_i^{\text{pe}} \\ &+ \sum_{j=1}^{N_p} \left(c_{j,i} \bar{u}_p^2 \bar{R}_1 \ln \left(\bar{\mathbf{I}} - \tanh^2(\rho_p(\nabla \psi_i^T(x) \hat{W}_{i,s})) \right) \right) \\ &- d_i^{\text{ep}} \bar{u}_e^2 \bar{R}_2 \ln \left(\bar{\mathbf{I}} - \tanh^2(\rho_e(\nabla \psi_i^T(x) \hat{W}_{i,s})) \right) \\ &+ \hat{W}_{i,s}^T \nabla \psi_i(x) \left(\sum_{j=1}^{N_p} (c_{j,i} \mathcal{F}_{j,i}(x)) \right) = 0. \end{aligned} \quad (42)$$

Define

$$\begin{aligned} K_{i,s} &\triangleq \nabla \psi_i(x) \left(\sum_{j=1}^{N_p} (c_{j,i} \mathcal{F}_{j,i}(x)) \right) \\ U_{N,i}^s &\triangleq \sum_{j=1}^{N_p} \left(c_{j,i} \bar{u}_p^2 \bar{R}_1 \ln \left(\bar{\mathbf{I}} - \tanh^2(\rho_p(\nabla \psi_i^T(x) \hat{W}_{i,s})) \right) \right) \\ W_{N,i}^s &\triangleq -d_i^{\text{ep}} \bar{u}_e^2 \bar{R}_2 \ln \left(\bar{\mathbf{I}} - \tanh^2(\rho_e(\nabla \psi_i^T(x) \hat{W}_{i,s})) \right). \end{aligned}$$

Then, we can further define

$$R_i^s \triangleq (\tilde{x}_i^{\text{pe}})^T Q \tilde{x}_i^{\text{pe}} + U_{N,i}^s + W_{N,i}^s \quad (43)$$

and thus, the residual error e_i^s can be rewritten as

$$e_i^s = R_i^s + \hat{W}_{i,s}^T K_{i,s}. \quad (44)$$

For obtaining the update rule of the weights of critic networks, the optimization objective is defined as

$$E_i^s = \frac{1}{2} (e_i^s)^T e_i^s. \quad (45)$$

The tuning law for the neural network weights $\hat{W}_{i,s}$ is given by

$$\begin{aligned} \dot{\hat{W}}_{i,s} &= -\alpha_i \left(\frac{\partial E_i^s}{\partial \hat{W}_{i,s}} \right)^T \\ &= -\alpha_i (R_i^s + \hat{W}_{i,s}^T K_{i,s}) K_{i,s}^T \end{aligned} \quad (46)$$

Algorithm 3: Off-Policy RL Algorithm.

Input: System states $\{x_j^p, x_i^e\}$ and their corresponding initial admissible control policy $\{\mu_{j,0}^p, \mu_{i,0}^e\}$. Initialize the error threshold ϵ , select an initial estimated weight vector $\hat{W}_{i,0}$. Set $s = 0$.

repeat

Based on the control pair $\{\mu_{j,s}^p, \mu_{i,s}^e\}$, solve (41) by using (46) to update the $\hat{W}_{i,s}$.

Update the control policy $\mu_{j,s}^p$ and $\mu_{i,s}^e$ simultaneously by

$$\begin{cases} \mu_{j,s}^p = -\bar{u}_p \tanh\left(\frac{1}{2\bar{u}_p} R_1^{-1} (g_j^p)^T \nabla \psi_i^T(x) \hat{W}_{i,s}\right) \\ \mu_{i,s}^e = -\bar{u}_e \tanh\left(\frac{1}{2\bar{u}_e} R_2^{-1} (g_i^e)^T \nabla \psi_i^T(x) \hat{W}_{i,s}\right). \end{cases}$$

Update the weighted adjacency matrix $\mathcal{A}_{pe}$ and $\mathcal{A}_{ep}$ simultaneously by using Algorithm 1.

Set $s = s + 1$.

until $|\hat{W}_{i,s} - \hat{W}_{i,s-1}| < \epsilon$.

Output: $\{\mu_{j,s}^p, \mu_{i,s}^e\}$.

where $\alpha_i > 0$ is the learning rate of the critic network.

According to Theorem 2 and [47], we make the following assumption.

Assumption 2: For any iteration instants $s > t_0$, there exist positive scalars $\vartheta_w, \vartheta_\psi, \lambda_\psi, \vartheta_\delta, \lambda_\delta$ and the time instant t_0 such that the following parameters are bounded [48]:

$$\begin{aligned} \|\hat{W}_{i,s}\| &< \vartheta_w, \|\psi_i(\cdot)\| < \vartheta_\psi, \|\nabla \psi_i(\cdot)\| < \lambda_\psi \\ \|\delta_s(\cdot)\| &< \vartheta_\delta, \|\nabla \delta_s(\cdot)\| < \lambda_\delta \end{aligned}$$

where the conditions on activation function can be satisfied by standard activation functions, e.g., by sigmoids, tanh, and other standard activation functions.

Then, the off-policy RL algorithm for MPE game is summarized as Algorithm 3.

Remark 4: The Algorithm 3 mainly includes the policy evaluation stage and policy improvement stage. To begin with, the correlative system data are collected. Then, the gradient descent algorithm is used to update the corresponding estimated weight. At last, the optimal control policies for MPE game with constrained inputs are iterated out. After the update of the weight adjacency matrix in each iteration, the corresponding edge weights in (41) will also be updated accordingly. Furthermore, to minimize the target function E_i^s , the estimated weight $\hat{W}_{i,s}$ can be calculated using (46). For the MPE game, we can calculate the optimal control policies with only one formula to replace (26) and (25) in Algorithm 2.

D. Stability Analysis

Assumption 3: Define $K_{i,s}$ is persistently existed. If at the time instance t_1 for an integer N_1 and $\check{\eta} > 0$, the following holds:

$$\sum_{s=t_1}^{t_1+N_1} K_{i,s} K_{i,s}^T \geq \check{\eta} \mathbf{I}_{i,s} \quad (47)$$

where $\mathbf{I}_{i,s}$ is the identity matrix.

In Theorems 2 and 3, the convergent property of Algorithm 2 is proved. In the following, the convergence of the weights of critic network based on off-policy RL algorithm is analyzed.

Before proceeding, the uniformly ultimately bounded (UUB) stability of the weight estimation error is defined as follows.

Theorem 4 (see [3]): Let the update method for critic networks be (46), define the weight estimation error $\tilde{W}_{i,s} = W_{i,s} - \hat{W}_{i,s}$. For each $i \in \{1, \dots, N_e\}$, the weight estimation error $\tilde{W}_{i,s}$ is said to be UUB under the following conditions.

We define a compact set $\Omega_3 \subset \mathbb{R}^h$ as the convergence set

$$\Omega_3 : \left\{ W_{i,s} \mid \|W_{i,s} - \hat{W}_{i,s}\| \leq \acute{m} b_i \right\}, \acute{m} > 1$$

where b_i is given by the formula

$$b_i = \sqrt{\frac{d^2 \left(\|W_{i,s}\|^2 \|K_{i,s}\|^2 + \|R_i^s\|^2 \right)}{2(d-1) \|K_{i,s}\|^2}}.$$

The condition for UUB is that for all $W_{i,s} \in \Omega_3$ there exists a bound b_i and a time instant $T(b_i, W_{i,s})$ such that $\|\tilde{W}_{i,s}\| \leq \acute{m} b_i$ for all $s \geq T$.

Proof: Choose the following Lyapunov function:

$$V_L \triangleq \frac{d}{2\eta_i} \tilde{W}_{i,s}^T \tilde{W}_{i,s} \quad (48)$$

with $d > 0$.

According to (46)

$$\begin{aligned} \dot{\tilde{W}}_{i,s} &= \alpha_i (R_i^s + (W_{i,s} - \tilde{W}_{i,s})^T K_{i,s}) K_{i,s}^T \\ &= -\alpha_i \tilde{W}_{i,s} K_{i,s} K_{i,s}^T + \alpha_i (R_i^s + W_{i,s} K_{i,s}) K_{i,s}^T. \end{aligned} \quad (49)$$

Take the derivative of V_L , one has

$$\begin{aligned} \dot{V}_L &= d(\tilde{W}_{i,s})^T \left((R_i^s + W_{i,s} K_{i,s}) K_{i,s}^T - \tilde{W}_{i,s} K_{i,s} K_{i,s}^T \right) \\ &= -d(\tilde{W}_{i,s})^T W_{i,s} K_{i,s} K_{i,s}^T + d(\tilde{W}_{i,s})^T \\ &\quad (R_i^s + W_{i,s} K_{i,s}) K_{i,s}^T \\ &\leq -d \|\tilde{W}_{i,s}\|^2 \|K_{i,s}\|^2 + d(\tilde{W}_{i,s})^T R_i^s (K_{i,s})^T \\ &\quad + d(\tilde{W}_{i,s})^T W_{i,s} K_{i,s} (K_{i,s})^T \\ &\leq -d \|\tilde{W}_{i,s}\|^2 \|K_{i,s}\|^2 + \frac{1}{2} \|\tilde{W}_{i,s}\|^2 \|K_{i,s}\|^2 \\ &\quad + \frac{d^2}{2} \|W_{i,s}\|^2 \|K_{i,s}\|^2 + \frac{1}{2} \|\tilde{W}_{i,s}\|^2 \|K_{i,s}\|^2 \\ &\quad + \frac{d^2}{2} \|R_i^s\|^2 \end{aligned} \quad (50)$$

i.e.,

$$\begin{aligned} \dot{V}_L &\leq -(d-1) \|\tilde{W}_{i,s}\|^2 \|K_{i,s}\|^2 + \frac{d^2}{2} \|W_{i,s}\|^2 \|K_{i,s}\|^2 \\ &\quad + \frac{d^2}{2} \|R_i^s\|^2. \end{aligned} \quad (51)$$

If $d > 1$, one has

$$\|\tilde{W}_{i,s}\|^2 > \frac{d^2 \left(\|W_{i,s}\|^2 \|K_{i,s}\|^2 + \|R_i^s\|^2 \right)}{2(d-1) \|K_{i,s}\|^2}. \quad (52)$$

Therefore, using Lyapunov theory, it can be concluded that the norm of the weight estimation error $\tilde{W}_{i,s}$ is bounded to within a small neighborhood of b_i . And thus, we can infer that $\hat{V}_i^s(x) \rightarrow V_i^s(x)$. ■

Remark 5: It has been proven that the magnitude of the weight estimation error is bounded within a finite region. To further

ensure the convergence of the weight estimation error $\tilde{W}_{i,s}$, such that the residual error magnitude remains within an acceptable range, the persistency of excitation assumption is provided [49], i.e., Assumption 3. For the implementation process, the persistency of excitation condition is satisfied by utilizing exploration noises added to the given control inputs. These noises introduce variability into the inputs, thereby maintaining the necessary level of excitation to ensure the desired convergence properties.

IV. SIMULATION

In this section, an MPE example based on the nonlinear aircraft system is presented to validate the effectiveness of the proposed off-policy RL control algorithm. Here, we present two simulation scenarios: multiple-pursuer-one-evader and multiple-pursuer-multiple-evader. In addition, we analyze the impact of saturation constraints, conduct comparative experiments on cohesion, and assess computational burden.

A. Nonlinear Aircraft Model

Consider an MPE game with agents described by (6) in $\mathbb{R}^3$. Letting system states (1) and (2) are both composed of $x_i^a, x_i^b, h_i, \check{v}_{a,i}, \check{v}_{b,i}$, and $v_{h,i}$. Control input both composed of u_i^a, u_i^b, u_i^h . According to [50], the dynamics of a simplified nonlinear aircraft model is designed as

$$f(x) = \begin{bmatrix} \check{v}_{a,i} \\ \check{v}_{b,i} \\ v_{h,i} \\ \frac{1}{\check{m}}(\check{T} \cos(\check{\alpha}) - \check{D}) \\ \frac{1}{\check{m}}(\check{L} + \check{T} \sin(\check{\alpha})) \\ \frac{1}{\check{m}}(\check{M} + \check{T} y_d) \end{bmatrix} \quad (53)$$

and

$$g(x) = \begin{bmatrix} 0_{3 \times 3} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 + 0.25 \sin(\check{v}_{a,i} x_i^b)^2 \end{bmatrix}. \quad (54)$$

Among these variables, x_i^a, x_i^b , and h_i represent positional information, while $\check{v}_{a,i}, \check{v}_{b,i}$, and $v_{h,i}$ represent velocity information. $\check{m}$ represents the mass of the aircraft. $\check{T}$ denotes the thrust generated by the engine. $\check{\alpha}$ signifies the angle of attack. y_d denotes the distance from the thrust application point to the aircraft's center of mass. $\check{D}, \check{L}$, and $\check{M}$ correspond to the drag force, lift force, and pitching moment acting on the aircraft, respectively. Calculation method of related parameters is given as

$$\check{T} = \bar{q} \check{S} C_T, \bar{q} = \frac{1}{2} \rho \check{V}^2, \rho = C_\rho \exp\left(\frac{-h_i}{24000}\right)$$

$$\check{V} = \sqrt{\check{v}_{a,i}^2 + \check{v}_{b,i}^2 + v_{h,i}^2}, C_T = C_\beta \check{\beta}$$

$$\check{\alpha} = \arctan\left(\frac{v_{h,i}}{\check{v}_{a,i}}\right), \check{\beta} = \arcsin\left(\frac{\check{v}_{b,i}}{\check{V}}\right)$$

where $\check{S}, \check{V}$, and C_T are the reference area, the true airspeed, and thrust coefficient, respectively. The relative value of the parameters is given in Table I.

TABLE I
PARAMETERS OF THE NONLINEAR AIRCRAFT SYSTEM

Parameter	$\check{m}$	$\check{S}$	C_ρ	C_β
Value	8920 kg	4m ²	0.00238	0.002576

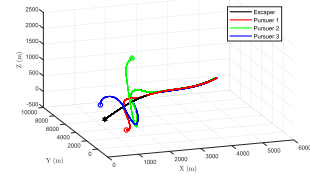


Fig. 1. Trajectory of the MPE game with three-pursuer-one-evader.

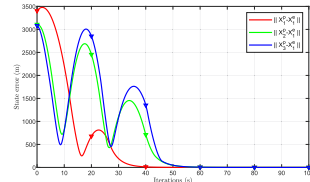


Fig. 2. State error between three pursuers and the evader.

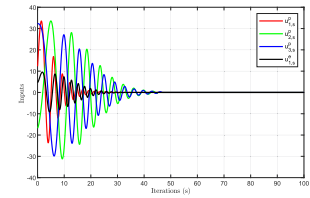


Fig. 3. Variations of the inputs between three pursuers and the evaders.

B. Scenario 1: Multiple-Pursuer-One-Evader

Consider an MPE game involving three pursuers and one evader in $\mathbb{R}^3$, where the dynamics of each agent are governed by system (53) and (54). This serves as a representative example of a more pursuers than evaders scenario. In the example, the initial conditions are $x_1^p = [600, -1900, 300, 78, 2, 18]^T$, $x_2^p = [1200, 500, 2200, 2, 20, 15]^T$, $x_3^p = [200, 600, 800, 5, 10, 125]^T$, $x_1^e = [500, 1500, 200; 50; 80; 10]$. Select $Q = I, R_1 = R_2 = I, \bar{u}_p = 35, \bar{u}_e = 15, \alpha_i = 0.01$ and the input constraint $\bar{u}_p = 25, \bar{u}_e = 15$. The expected displacement is given by $r_{1,1}^{pe} = [50, 10, 0, 0, 0, 0]$, $r_{2,1}^{pe} = [10, 50, 0, 0, 0, 0]$, and $r_{3,1}^{pe} = [-10, 0, -50, 0, 0, 0]$. Those varying expected displacements aid the pursuers in effectively surrounding the evader. The activation functions $\psi_i(\cdot)$ are selected as sigmoid functions. Then, the capture results of the three-pursuer-one-evader trajectory are shown in Fig. 1. From Fig. 1, it is obvious that at the beginning of the simulation, there is a considerable distance between the pursuers and the evader. After a brief adjustment, the distances between the three pursuers and the evader quickly reduce to a very small value and are maintained thereafter. And the state error between the pursuers and the evader is presented in Fig. 2. Fig. 3 shows the control inputs $u_{i,s}^p$ and $u_{i,s}^e$ of the pursuers and the evader, respectively. From Fig. 3, it can be observed that the constraints on saturated inputs are effectively implemented.

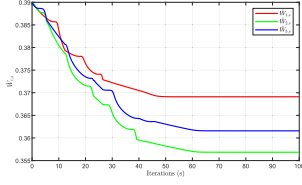


Fig. 4. Convergence curves of weight estimate value $\hat{W}_{i,s}$.

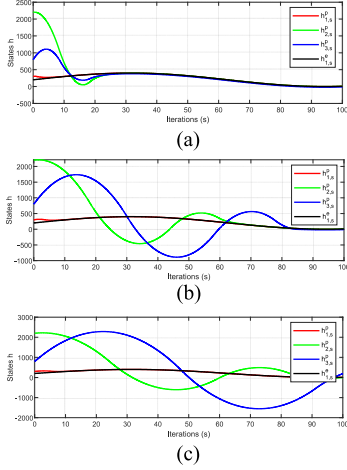


Fig. 5. Comparison of h_i under different saturation constraints. (a) $\bar{u}_p > \bar{u}_e$. (b) $\bar{u}_p = \bar{u}_e$. (c) $\bar{u}_p < \bar{u}_e$.

From Fig. 4, we can observe the convergence process of the critic network weights $\hat{W}_{i,s}$ updated by (46). Thus, the approximate value functions converge to the optimal ones. Here, we calculate and plot the square of the Euclidean norm of the $\hat{W}_{i,s}$ vector. To be specific, the weight vector $\hat{W}_{1,s}$ converges to $[0.1997, 0.1453, 0.1995, 0.3005, 0.2966, 0.2998]^T$, the weight vector $\hat{W}_{2,s}$ converges to $[0.1996, 0.1286, 0.1956, 0.3002, 0.2871, 0.2992]^T$, and the weight vector $\hat{W}_{3,s}$ converges to $[0.1940, 0.1342, 0.2002, 0.3014, 0.2909, 0.3004]^T$ at the end of the simulation.

To compare and validate the impact of different saturation constraints $\bar{u}_p$ and $\bar{u}_e$ on the algorithm's results, the algorithm is evaluated under three conditions: $\bar{u}_p > \bar{u}_e$, $\bar{u}_p = \bar{u}_e$, and $\bar{u}_p < \bar{u}_e$. With all other conditions held constant, the state tracking of pursuers and evaders under different saturation constraints is depicted in Fig. 5. In Fig. 5(a), $\bar{u}_p = 25, \bar{u}_e = 20$, in Fig. 5(b), $\bar{u}_p = \bar{u}_e = 20$, and in Fig. 5(c), $\bar{u}_p = 15, \bar{u}_e = 20$. From the figure, it can be observed that as $\bar{u}_p$ decreases relative to $\bar{u}_e$, the time required for the state error to diminish also increases.

C. Scenario 2: Multiple-Pursuer-Multiple-Evader

Consider the following scenario with three pursuers and three evader in $\mathbb{R}^3$. This serves as an example of an MPE game with multiple pursuers and multiple evaders. The initial states of pursuers and evaders are $x_1^p = [263, -2380, 800, 123, 12, 18]^T$, $x_2^p = [-380, 3100, -200, 12, -40, 15]^T$, $x_3^p = [2500, -3280, 400, -15, 30, 15]^T$, $x_1^e = [1037, 360, 100, 80, 40, 0]^T$, $x_2^e = [521, 1030, 34, 0, 80, 1.6]^T$, $x_3^e = [1816, -1887, -90, 40, 65, -0.1]^T$, $\bar{u}_p = 20, \bar{u}_e = 20$. The expected displacement $r_{j,i}^{pe} = [100, 50,$

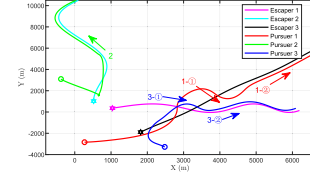


Fig. 6. MPE Game's Trajectory using our algorithm, three-pursuer-three-evader scenario.

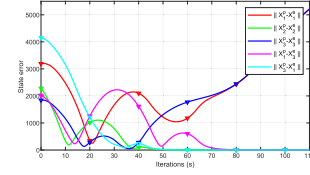


Fig. 7. State errors between pursuers and their respective old/new targets in our algorithm.

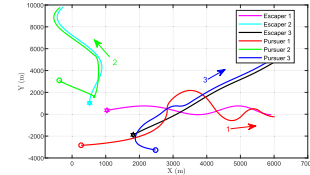


Fig. 8. Trajectory of the MPE game in a three-pursuer-three-evader setting using [26].

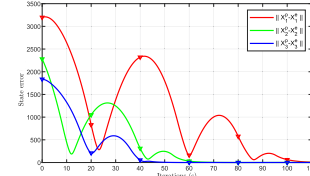


Fig. 9. State errors between three pursuers and their respective target using [26].

$50, 0, 0, 0]$. The activation functions $\psi_i(\cdot)$ are selected as sigmoid functions. The swap threshold $\rho = 50$.

The results of the three-pursuer-three-evader trajectory are shown in Fig. 6. The corresponding state error between the pursuers and their respective old/new target evader is presented in Fig. 7. It can be observed from the trajectory that the targets of the pursuers undergo changes during the process under the dynamic target graph algorithm. The original and the reassigned weighted adjacency matrices, denoted as $\mathcal{A}_{pe}$ and $\mathcal{A}'_{pe}$ are shown as

$$\mathcal{A}_{pe} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathcal{A}'_{pe} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

As a comparative experiment, under the same conditions, we have implemented MPE experiments using the target topology in [26]. The corresponding trajectory and state error are shown in Figs. 8 and 9, respectively. In existing works [17], [18], [39] on the MPE problem with fixed topologies, different control strategies are employed, but all use graphs to model agent

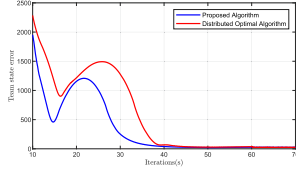


Fig. 10. Team cohesion comparison, proposed method versus [26].

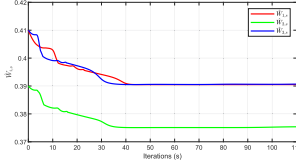
Fig. 11. Convergence curves of weight estimate value $\hat{W}_{i,s}$.

TABLE II
COMPARATIVE ANALYSIS OF COMPUTATION BURDEN IN V-SNAC AND AC NETWORKS

Network	Scenario	Time per step (ms)
AC [39]	Three-pursuer-one-evader	36.73
	Three-pursuer-three-evader	41.95
V-SNAC	Three-pursuer-one-evader	25.34
	Three-pursuer-three-evader	29.17

Bold values indicate the reduced time spent on computing each step with the V-SNAC.

interactions. According to Theorem 1, the dynamic target graph method introduced here can further reduce the team state error and boost team cohesion in these fixed-topology approaches. The team state error calculated by (7) is utilized to compare team cohesion. Smaller team state error indicating stronger cohesion among the pursuer team. It can be observed from Fig. 10 that our method exhibits faster reduction in team state error and demonstrates better team cohesion.

The convergence curves of $\hat{W}_{i,s}$ are displayed in Fig. 11. Thus, the approximate value functions converge to the optimal ones within a small bound. To elaborate, the weight vector $\hat{W}_{1,s}$ converges to $[0.1468, 0.1883, 0.1986, 0.3042, 0.2033, 0.4002]^T$, the weight vector $\hat{W}_{2,s}$ converges to $[0.1943, 0.1545, 0.1995, 0.3012, 0.3051, 0.3001]^T$, and the weight vector $\hat{W}_{3,s}$ converges to $[0.1372, 0.1930, 0.1985, 0.3094, 0.1978, 0.4003]^T$ at the end of the simulation. This also confirms the stability of the control policy.

Because the V-SNAC structure adopted by our method reduces the number of neural networks, we have compared its computational time with the traditional AC structure [39]. Table II records the computational time required for a single step of control under the two network structures, denoted as “time per step.” The comparative analysis reveals that the V-SNAC framework achieves a reduction of approximately 31% in computation time per step, which demonstrate the efficiency of the V-SNAC structure and provides a possibility for faster decision-making in the context of the MPE game.

Remark 6: The abovementioned two simulation scenarios represent two categories of typical cases. Under the assumption that $N_p \geq N_e$, the MPE game can be classified into

the aforementioned combinations of multi-pursuer-multi-evader and multi-pursuer-single-evader, which can be solved in the same manner. Therefore, the simulation examples confirm the effectiveness of the designed off-policy RL Algorithm with dynamic target graph and V-SNAC structure. Furthermore, it also shows that the off-policy RL scheme is applicable to the nonlinear MPE games with constrained inputs.

V. CONCLUSION

An approximate optimal control strategy based on reinforcement learning has been designed and applied to the multiagent PE game in this article. The relationships between agents are represented using graph models. The MPE game is constrained by actuator saturation and nonlinear system dynamics. Initially, using the proposed approximate HJI equation, we present an on-policy PI algorithm with convergence analysis. Then, to approximate the value functions, an off-policy RL method is introduced. This algorithm efficiently solves the approximate HJI equation and provides solutions under multiagent conditions. We utilize a network based on the V-SNAC structure to implement this off-policy RL method. Furthermore, it is noteworthy that the proof of the network system’s UUB is provided. Finally, we conduct simulations in two distinct scenarios: multiple-pursuer-one-evader and multiple-pursuer-multiple-evader. Comparative analyzes on cohesion and computational burden are employed to verify the effectiveness. In future research, attention can be focused on enhancing the collaboration among evaders, as well as addressing the problem of MPE in situations with limited state perception and communication.

REFERENCES

- [1] R. Zhang, Q. Zong, X. Zhang, L. Dou, and B. Tian, “Game of drones: Multi-UAV pursuit-evasion game with online motion planning by deep reinforcement learning,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7900–7909, Oct. 2023.
- [2] H. Gong, S. Gong, and J. Li, “Pursuit–evasion game for satellites based on continuous thrust reachable domain,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 6, pp. 4626–4637, Dec. 2020.
- [3] F. Lewis, S. Jagannathan, and A. Yesildirak, *Neural Network Control of Robot Manipulators and Non-Linear Systems* (Systems and Control). New York, NY, USA: Taylor & Francis, 1998.
- [4] C. Hong, I. Jeong, L. F. Vecchietti, D. Har, and J.-H. Kim, “AI world cup: Robot-soccer-based competitions,” *IEEE Trans. Games*, vol. 13, no. 4, pp. 330–341, Dec. 2021.
- [5] J. Chen, W. Zha, Z. Peng, and D. Gu, “Multi-player pursuit–evasion games with one superior evader,” *Automatica*, vol. 71, pp. 24–32, 2016.
- [6] J. Wang, C. Wang, Y. Wei, and C. Zhang, “Neuroadaptive sliding mode formation control of autonomous underwater vehicles with uncertain dynamics,” *IEEE Syst. J.*, vol. 14, no. 3, pp. 3325–3333, Sep. 2020.
- [7] R. Isaacs, *Differential Games: A Mathematical Theory With Applications to Warfare and Pursuit, Control and Optimization*. North Chelmsford, MA, USA: Courier Corporation, 1999.
- [8] J. S. Jang and C. Tomlin, “Control strategies in multi-player pursuit and evasion game,” in *Proc. Amer. Inst. Aeronaut. Astronaut. Guid., Navigat., Control Conf. Exhib.*, 2005, Art. no. 6239.
- [9] A. Jagat and A. J. Sinclair, “Nonlinear control for spacecraft pursuit-evasion game using the state-dependent Riccati equation method,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 6, pp. 3032–3042, Dec. 2017.
- [10] W. Lin, Z. Qu, and M. A. Simaan, “Nash strategies for pursuit-evasion differential games involving limited observations,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 2, pp. 1347–1356, Apr. 2015.
- [11] V. G. Lopez, F. L. Lewis, Y. Wan, E. N. Sanchez, and L. Fan, “Solutions for multiagent pursuit-evasion games on communication graphs: Finite-time capture and asymptotic behaviors,” *IEEE Trans. Autom. Control*, vol. 65, no. 5, pp. 1911–1923, May 2020.

- [12] Y. Xu, H. Yang, B. Jiang, and M. M. Polycarpou, "Multiplayer pursuit-evasion differential games with malicious pursuers," *IEEE Trans. Autom. Control*, vol. 67, no. 9, pp. 4939–4946, Sep. 2022.
- [13] A. Jenabzadeh and B. Safarinejadian, "Distributed tracking of nonholonomic targets over multiagent systems," *IEEE Syst. J.*, vol. 13, no. 2, pp. 1678–1681, Jun. 2019.
- [14] D. K. Kim et al., "A policy gradient algorithm for learning to learn in multiagent reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 5541–5550.
- [15] T. T. Nguyen, N. D. Nguyen, and S. Nahavandi, "Deep reinforcement learning for multiagent systems: A review of challenges, solutions, and applications," *IEEE Trans. Cybern.*, vol. 50, no. 9, pp. 3826–3839, Sep. 2020.
- [16] K. G. Vamvoudakis, F. L. Lewis, and G. R. Hudas, "Multi-agent differential graphical games: Online adaptive learning solution for synchronization with optimality," *Automatica*, vol. 48, no. 8, pp. 1598–1611, 2012.
- [17] H. Zhang, W. Zhao, H. Ge, X. Xie, and D. Yue, "Distributed model-free optimal control for multiagent pursuit-evasion differential games," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 4, pp. 3800–3811, Aug. 2024.
- [18] R. Sarkar, D. Patil, A. K. Mulla, and I. N. Kar, "Finite-time consensus tracking of multi-agent systems using time-fuel optimal pursuit evasion," *IEEE Control Syst. Lett.*, vol. 6, pp. 962–967, 2022.
- [19] M. Liu, Y. Wan, V. G. Lopez, F. L. Lewis, G. A. Hewer, and K. Estabridis, "Differential graphical game with distributed global Nash solution," *IEEE Trans. Control Netw. Syst.*, vol. 8, no. 3, pp. 1371–1382, Sep. 2021.
- [20] D. Cappelletto and T. Mylvaganam, "Distributed differential games for control of multi-agent systems," *IEEE Trans. Control Netw. Syst.*, vol. 9, no. 2, pp. 635–646, Jun. 2022.
- [21] D. Yu, C. P. Chen, C.-E. Ren, and S. Sui, "Swarm control for self-organized system with fixed and switching topology," *IEEE Trans. Cybern.*, vol. 50, no. 10, pp. 4481–4494, Oct. 2020.
- [22] X. Liu, X. Wu, Y. Xie, and B. Guo, "Time-varying formation control with transient performance for multiagent systems under jointly connected topologies," *IEEE Syst. J.*, vol. 18, no. 1, pp. 497–500, Mar. 2024.
- [23] W. Zou, C. K. Ahn, and Z. Xiang, "Event-triggered consensus tracking control of stochastic nonlinear multiagent systems," *IEEE Syst. J.*, vol. 13, no. 4, pp. 4051–4059, Dec. 2019.
- [24] J. Zhuang, S. Peng, and Y. Wang, "Leader-following consensus of discrete-time stochastic nonlinear multiagent systems under fixed and switching topologies via impulsive control," *IEEE Syst. J.*, vol. 16, no. 4, pp. 6021–6030, Dec. 2022.
- [25] D. Liu, H. Li, and D. Wang, "Online synchronous approximate optimal learning algorithm for multi-player non-zero-sum games with unknown dynamics," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 44, no. 8, pp. 1015–1027, Aug. 2014.
- [26] P. Zhou and B. M. Chen, "Distributed optimal solutions for multiagent pursuit-evasion games for capture and formation control," *IEEE Trans. Ind. Electron.*, vol. 71, no. 5, pp. 5224–5234, May 2024.
- [27] Q. Wei, D. Liu, Q. Lin, and R. Song, "Discrete-time optimal control via local policy iteration adaptive dynamic programming," *IEEE Trans. Cybern.*, vol. 47, no. 10, pp. 3367–3379, Oct. 2017.
- [28] H. Xu, D. Yu, Z. Wang, K. H. Cheong, and C. P. Chen, "Nonsingular predefined time adaptive dynamic surface control for quantized nonlinear systems," *IEEE Trans. Syst., Man, Cybern. Syst.*, early access, Jul. 2024. doi: [10.1109/TSMC.2024.3407150](https://doi.org/10.1109/TSMC.2024.3407150).
- [29] P. Zhou and B. M. Chen, "Distributed optimal solutions for multiagent pursuit-evasion games," in *Proc. IEEE 62nd Conf. Decis. Control*, 2023, pp. 6424–6429.
- [30] H. Xiong and Y. Zhang, "Reinforcement learning-based formation-surrounding control for multiple quadrotor UAVs pursuit-evasion games," *ISA Trans.*, vol. 145, pp. 205–224, 2024.
- [31] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 6382–6393.
- [32] J. Ding and S. Balakrishnan, "A single network approximate dynamic programming based constrained optimal controller for nonlinear systems with uncertainties," in *Proc. IEEE 49th Conf. Decis. Control*, 2010, pp. 3054–3059.
- [33] Y. Yuan, P. Zhang, and X. Li, "Synchronous fault-tolerant near-optimal control for discrete-time nonlinear PE game," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 10, pp. 4432–4444, Oct. 2021.
- [34] J. Long, D. Yu, G. Wen, L. Li, Z. Wang, and C. P. Chen, "Game-based backstepping design for strict-feedback nonlinear multi-agent systems based on reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 817–830, Jan. 2024.
- [35] D. Yu, J. Long, C. P. Chen, and Z. Wang, "Adaptive swarm control within saturated input based on nonlinear coupling degree," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 52, no. 8, pp. 4900–4911, Aug. 2022.
- [36] T. Yang, Z. Meng, D. V. Dimarogonas, and K. H. Johansson, "Global consensus for discrete-time multi-agent systems with input saturation constraints," *Automatica*, vol. 50, no. 2, pp. 499–506, 2014.
- [37] H. Xu, D. Yu, and Y.-J. Liu, "Observer-based fuzzy adaptive predefined time control for uncertain nonlinear systems with full-state error constraints," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 3, pp. 1370–1382, Mar. 2024.
- [38] H. Zhang, Z. Li, Z. Qu, and F. L. Lewis, "On constructing lyapunov functions for multi-agent systems," *Automatica*, vol. 58, pp. 39–42, 2015.
- [39] X. Dong, H. Zhang, and Z. Ming, "Adaptive optimal control via Q-learning for multi-agent pursuit-evasion games," *IEEE Trans. Circuits Syst. II: Exp. Briefs*, vol. 71, no. 6, pp. 3056–3060, Jun. 2024.
- [40] R. Song, F. L. Lewis, and Q. Wei, "Off-policy integral reinforcement learning method to solve nonlinear continuous-time multiplayer nonzero-sum games," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 3, pp. 704–713, Mar. 2017.
- [41] H. P. Geering, *Optimal Control With Engineering Applications*, vol. 113. Berlin, Germany: Springer, 2007.
- [42] H. Modares, F. L. Lewis, and Z.-P. Jiang, " H_∞ tracking control of completely unknown continuous-time systems via off-policy reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 10, pp. 2550–2562, Oct. 2015.
- [43] H.-N. Wu and B. Luo, "Neural network based online simultaneous policy update algorithm for solving the HJI equation in nonlinear H_∞ control," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 12, pp. 1884–1895, Dec. 2012.
- [44] K. Hoffman, *Banach Spaces of Analytic Functions*. New York, NY, USA: Dover, 2014.
- [45] L. Boron and E. Zeidler, *Nonlinear Functional Analysis and Its Applications: III: Variational Methods and Optimization*. New York, NY, USA: Springer, 2013.
- [46] E. Zeidler, *Nonlinear Functional Analysis and Its Applications: I: Fixed-Point Theorems* (Nonlinear Functional Analysis and Its Applications). Berlin, Germany: Springer, 1985.
- [47] J. Sarangapani, *Neural Network Control of Nonlinear Discrete-Time Systems* (Automation and Control Engineering). Boca Raton, FL, USA: CRC Press, 2018.
- [48] K. Hornik, M. Stinchcombe, and H. White, "Universal approximation of an unknown mapping and its derivatives using multilayer feedforward networks," *Neural Netw.*, vol. 3, no. 5, pp. 551–560, 1990.
- [49] Y. Huo, D. Wang, J. Qiao, and M. Li, "Off-policy model-free learning for multi-player non-zero-sum games with constrained inputs," *IEEE Trans. Circuits Syst. I: Regular Papers*, vol. 70, no. 2, pp. 910–920, Feb. 2023.
- [50] B. Xu, F. Sun, C. Yang, D. Gao, and J. Ren, "Adaptive discrete-time controller design with neural network for hypersonic flight vehicle via back-stepping," *Int. J. Control*, vol. 84, no. 9, pp. 1543–1552, 2011.